

- 2 (a)** In eukaryotes, the cells of plants and photosynthetic protists contain chloroplasts.

Describe the relationship between the function of a chloroplast and its structure.

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- (b)** Outline the main features of cyclic photophosphorylation.

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- (c) Red algae are aquatic protocists that are multicellular. The cells of red algae have chloroplasts containing photosynthetic pigments. Many red algae live in deep water.

Two of the accessory pigments of red algae chloroplasts are:

- phycoerythrin (appears red), often present in large concentrations
- phycocyanin (appears blue).

The first few metres of water nearest the surface absorb the red wavelengths of light. If the water also contains particles of organic material it absorbs blue wavelengths.

Fig. 2.1 shows absorption spectra of some pigments in red algae chloroplasts.

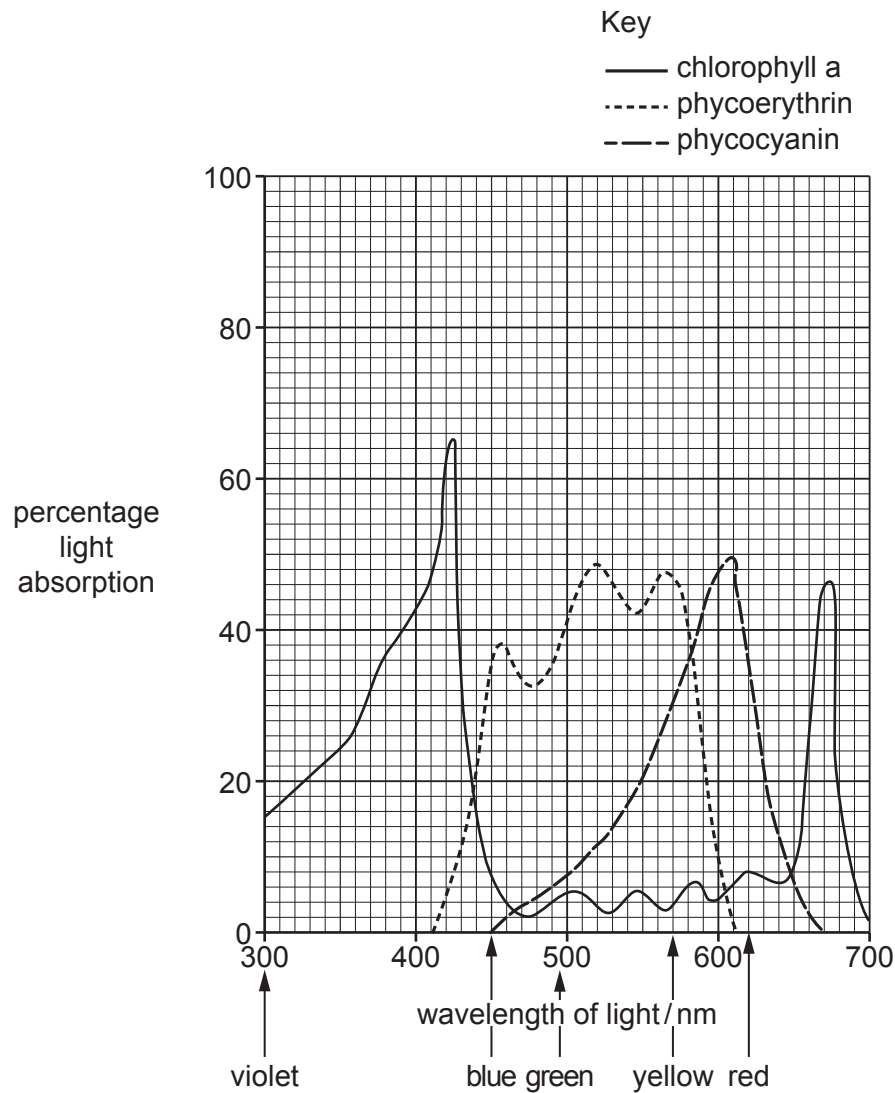


Fig. 2.1

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